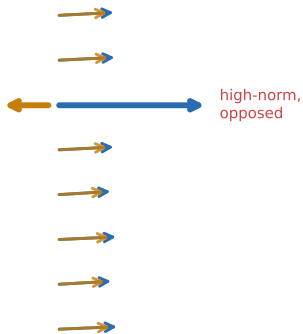


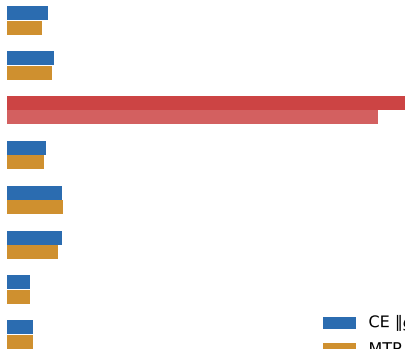
Aligned per row, but a high-norm opposed minority drags the aggregate negative; the cosine misreads it

Per row: direction
99.7% aligned; a few opposed



one row per vocab token

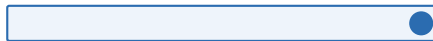
Per row: magnitude, same rows
(support overlaps, $\cos \approx 0.98$)



CE $\|g\|$
MTP $\|g\|$

Two metrics, two readings

per-row cosine (typical row)



aggregate cosine (mass-weighted)



the opposed minority dominates the sum